

Control-impact architecture of the human structural connectome: degree dominance, a small near-universal residual, and a cross-scale architectural comparison with the fly connectome

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Abstract

Background. Network communication in brain connectomes is dominated by node degree, but whether any degree-independent component of node-level control impact exists—and whether it is anatomically structured—remains unresolved, because most node-removal analyses are group-average, lack matched controls, and lack null-model arbitration. **Gap.** No prior study, to our knowledge, has combined exact per-node removal-based control impact with per-subject population-scale statistics, degree/strength matching, degree-preserving null ensembles, and a pre-registered cross-scale comparison. **Methods.** We computed the exact Control Impact Score, $CIS(i) = (E(G) - E(G - i))/E(G)$, for all 456 nodes in each of 801 QC-pass subjects from the AOMIC-ID1000 structural-connectome resource ($N = 900$ acquired; 456-node 4S456 parcellation; SIFT streamline weights; 15% proportional threshold), with degree-matched controls ($\pm 10\%$), 100 degree-preserving (Maslov–Sneppen) nulls per graph in two batteries (100×100 full-CIS; 200×100 top-50), BH-FDR node-level inference, system-enrichment tests with two control levels, and a 9-configuration robustness ladder. **Results.** Top-50 CIS concentration exceeded its degree-preserving null in 200/200 subjects (median $z = 16.36$; two-sided sign $p = 1.24 \times 10^{-60}$). CIS was strongly degree-related (median $\rho = 0.943$). A small degree-independent residual was present in 778/801 subjects (sign $p = 2.6 \times 10^{-197}$; Cliff’s $\delta = 0.100$, across-subject IQR $[0.078, 0.129]$; median Wilcoxon $p = 0.167$), and 43/456 nodes survived FDR ($q < .05$; max Stouffer $z = 23.30$). Compared with the frozen fly (FAFB v783) study ($\delta = 0.0979$, itself not surviving its null ensemble), the human architecture is concordant in effect size and degree dominance, while the residual’s anatomical identity diverges: visual systems hold 2% of the human top 50 versus 80% visual-centrifugal in the fly; the pre-registered system-enrichment gate was negative. **Interpretation.** Control-impact architecture—a dominant degree component plus a small, near-universal degree-independent residual—shows cross-scale architectural similarity while the residual’s anatomical identity diverges. No new biological mechanism, homology, or system-specific claim is made.

Keywords: structural connectome; network neuroscience; control impact; node removal; degree-preserving null model; population analysis; cross-species comparison; global efficiency.

1 Introduction

Node-removal and efficiency-based analyses of human structural connectomes are classical tools for quantifying node-level influence [1, 2], and hub and rich-club organization of the human connectome is well documented [3]. A persistent confound, however, is node degree: raw comparisons of node importance are dominated by how strongly a node is connected, so that observed “importance” can reflect degree alone. Matched controls and degree-preserving null models are both required to separate these effects [4]; the decisive lesson of the present project’s frozen fly-connectome predecessor is that matched-pair significance can dissolve entirely under degree-preserving null arbitration.

Whether a *degree-independent* component of control impact exists at the single-subject level, is reproducible across a population, and is anatomically structured cannot be answered by group-average designs: averaging across subjects destroys the per-subject variability that such a component would need to be

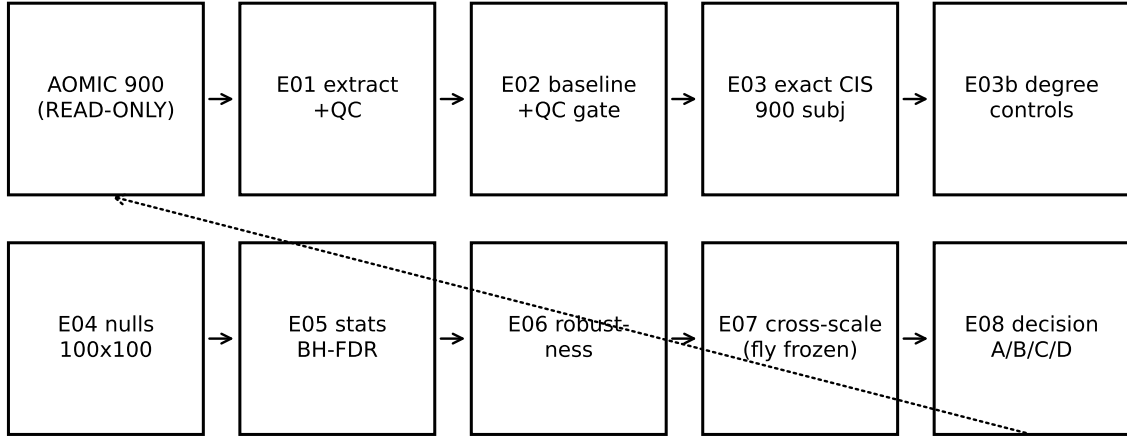


Figure 1: Study design and pipeline. From the population-scale AOMIC-ID1000 structural-connectome resource (900 subjects acquired; 801 QC-pass on frozen, CIS-blind flags), exact per-node Control Impact Scores were computed for all 456 nodes in every subject, followed by degree-matched controls ($\pm 10\%$ total degree), degree-preserving (Maslov–Sneppen) null ensembles in two batteries, BH-FDR node-level inference, two-level system-enrichment tests, a 9-configuration robustness ladder, and a pre-registered cross-scale comparison with the frozen fly (FAFB v783) study.

detected against. Network-controllability approaches [5, 6] address influence from a complementary, dynamical perspective but do not provide per-subject removal-based rankings with null arbitration; simulated-lesion studies either operate on group averages [1, 2, 7] or use matched-mass rather than degree-preserving nulls in an ageing framing [8]. Cross-species connectome comparisons [9, 10] and the Amsterdam Open MRI Collection [11] supply the comparative and data foundations for the present design.

Research question. Can removal-based control impact identify structurally influential nodes in individual human structural connectomes at population scale, and does a degree-independent component of that impact exist—and is it anatomically structured?

Hypotheses (pre-registered in PROTOCOL_FREEZE.md before any computation). **H1:** top-50 CIS concentration exceeds degree-preserving null expectations. **H2:** a degree-independent residual remains after degree/strength adjustment. **H3:** the residual is robustly system-specific. **H4:** comparable architecture appears across fly and human scales.

Contribution. To our knowledge, no prior study has combined per-node removal-based network-impact analysis, population-scale individual human structural connectomes, explicit degree-controlled residualization, degree-preserving null-model arbitration, FDR-controlled residual inference, and a pre-specified cross-species comparison with the *Drosophila* connectome in a single analytical framework. The novelty lies primarily in this integration and the resulting cross-scale empirical comparison, not in any individual component: each method is established [1, 3, 5, 6, 4].

2 Methods

2.1 Study overview

Figure 1 summarizes the pipeline. All analyses were pre-registered and frozen before outcome evaluation (PROTOCOL_FREEZE.md, STUDY_DESIGN.md); every number in this manuscript traces to a frozen artifact, and an independent stdlib-only audit recomputes all documented values from those artifacts (46/46 checks; `verify_final_numbers.py`).

Table 1: Primary cohort and baseline network properties. Structural connectomes were extracted for 900 subjects; 801 pass the pre-registered, CIS-blind property-based QC (flags retained, never deleted). Baseline statistics are for the primary configuration (4S456Parcels, sift_radius2_count weights, 15% proportional threshold). E0: global efficiency; k : mean degree. Values are frozen artifact outputs (table_01_subject_cohort.csv, table_03_qc_summary.csv, table_02_baseline_network.csv).

Quantity	Value	Source
Subjects acquired	900	frozen manifest
QC-pass primary cohort	801 (0.89)	E02 property QC
Flags retained (isolated-node / strength-IQR)	94 / 5	qc_primary.csv
Parcellation (nodes N)	4S456Parcels ($N = 456$)	atlas manifest
Weight variant	sift_radius2_count_connectivity	frozen config
Proportional threshold c	0.15 ($k = 15,561$ edges)	E01
Mean degree $\langle k \rangle$	68.2	baseline artifact
Global efficiency E_0 (median)	0.5468 [0.5441–0.5497]	baseline artifact

2.2 Dataset and cohort

Structural connectomes were drawn from the AOMIC-ID1000 derivative (Zenodo 19796783; CC-BY-4.0) [11]: 900 subjects $\times$ 7 atlases $\times$ 4 weight variants, checksum-verified on acquisition (10 archives, 7,067,175,601 bytes, byte-exact against the Zenodo API). The primary configuration is 4S456Parcels ($N = 456$ nodes) with sift_radius2_count_connectivity weights at a 15% proportional threshold ($k = 15,561$ edges). Cohort membership was frozen from pre-registered, CIS-blind, property-based QC (E02): 801/900 subjects pass (94 isolated-node and 5 strength-IQR flags; flagged subjects retained, never deleted). Thresholding keeps the $k = c N(N - 1)/2$ largest off-diagonal weights per graph (raw densities 0.45–0.88 preclude unthresholded path-length statistics); the cost ladder $\{0.10, 0.20, 0.25\}$ is used for robustness. Population baseline statistics are given in Table 1.

2.3 Control Impact Score

For each subject graph G and node i , the Control Impact Score is the fraction of baseline global efficiency lost on exact removal of i :

$$\text{CIS}(i) = \frac{E(G) - E(G - i)}{E(G)}, \quad E(G) = \frac{1}{N(N - 1)} \sum_{\substack{u,v=1 \\ u \neq v}}^N \frac{1}{d(u, v)}, \quad (1)$$

where $d(u, v)$ is the graph shortest-path distance in the binary undirected thresholded graph and $E(G - i)$ is evaluated on the induced $(N - 1)$ -node subgraph with $(N - 1)(N - 2)$ normalization. All 456 per-node values are computed exactly (Dijkstra; scipy) per subject. The reference implementation was validated against a naive implementation to 6.1×10^{-16} relative error; a boolean-matrix fast path used *only* for null graphs is equivalence-gated at 10^{-9} (max deviation 4×10^{-16}). Figure 2 summarizes the measurement, control, and arbitration logic; Fig. 3(A) shows the population-mean CIS architecture (maximum population-mean CIS 0.00462 at node 414; frozen value 0.0046246).

2.4 Degree/strength controls and residual analysis

Each top-decile CIS node is paired 1:1 with the nearest node within $\pm 10\%$ total degree (greedy, no replacement; nearest-50 fallback flagged), yielding 36,846 degree-matched pairs. The degree-independent residual is

$$r(i) = \text{CIS}(i) - \text{CIS}(c_{\text{match}}(i)), \quad (2)$$

summarized per subject by Cliff’s delta $\delta = \frac{\#\{(i, c): r(i) > 0\}}{N_{\text{pairs}}} - \frac{\#\{(i, c): r(i) < 0\}}{N_{\text{pairs}}}$. The strength association is reported alongside (median $\rho(\text{CIS}, \text{strength}) = 0.486$ vs. $\rho(\text{CIS}, \text{degree}) = 0.943$; Fig. 3B).

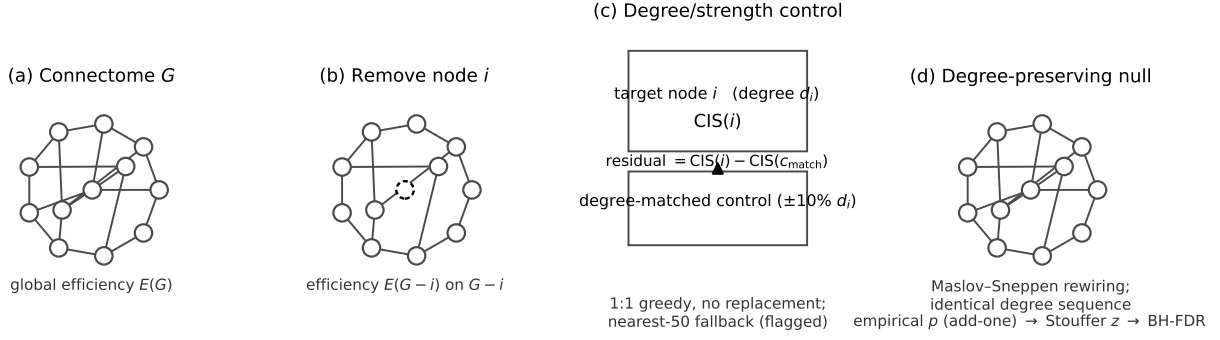


Figure 2: CIS measurement, control, and arbitration (schematic; no data plotted). (a) Baseline thresholded connectome G with global efficiency $E(G)$ (Eq. 1). (b) Exact removal of node i yields $E(G-i)$ and $\text{CIS}(i)$. (c) Degree/strength control: each top-decile target node is paired 1:1 with a $\pm 10\%$ -degree-matched control (greedy, no replacement; nearest-50 fallback flagged per pair); the residual is $\text{CIS}(i) - \text{CIS}(c_{\text{match}})$. (d) Arbitration: Maslov-Sneppen rewiring preserves the exact per-node degree sequence; per-node empirical p -values (add-one) feed Stouffer combination and BH-FDR, and per-subject top-50 concentration is compared against its own null ensemble.

2.5 Degree-preserving nulls

Null graphs are generated by undirected Maslov-Sneppen double-edge swaps [4] preserving the exact per-node degree sequence, with $n_{\text{swaps}} = 10 |E|$ attempted swaps per null, rejection of self-loops, duplicates, and degenerate picks, and exact per-null degree verification (10,000/10,000 exact; 0 self-loops; swap acceptance 49.4%). Two batteries: **battery A** (100 subjects $\times$ 100 nulls $\times$ 456-node full-CIS recomputation) for per-node inference; and **battery B** (200 subjects $\times$ 100 nulls of the top-50 median CIS; 100 fresh nulls plus 100 reused from battery A) for per-subject concentration. Seeds are frozen (nulls $100+i$ per graph; subject samples nested under a single RNG with seed 20260922; all recorded in null_manifest_battery_a.csv).

2.6 Statistical inference

Battery B (H1): per-subject z against the null distribution of the top-50 median CIS; population two-sided sign test. **Battery A (R2a):** per-node one-sided empirical p with the add-one correction $p = (\#\{\text{nulls} \geq \text{obs}\} + 1) / (100 + 1)$, Stouffer combination $z_j = \sum_s z_{sj} / \sqrt{n_s}$, and Benjamini-Hochberg FDR at $q = .05$ (single stratum). Censored nodes (all nulls $\geq$ observed; 413 of 456) carry $p = 1$ and $z = -\infty$ by construction and are counted as censored, not hidden. **System enrichment (H3/R2b):** control A = node-label shuffle (10,000 permutations); control B = degree-matched peer pools (10,000 resamples); z and empirical p with add-one floor. **Rank stability:** 100 seeded half-splits, Spearman correlation of mean-CIS ranks. **Effect sizes:** Cliff's δ , reported as the median across subjects with the across-subject IQR.

2.7 Cross-species comparison (pre-registered)

The human analysis is compared with the frozen fly study (FAFB v783, 139,255 reconstructed neurons) using normalized comparators only, with explicit discipline tags (DIRECT / NORMALIZED / QUALITATIVE; Table 4). Human and fly nodes are not treated as homologous structures; the comparison operates at the level of network architecture and statistical organization across a $\sim 10^3$ -fold difference in analyzed node count. Absolute CIS values are never compared across species.

3 Results

3.1 Population CIS architecture and its degree relation

CIS was computed for all 456 nodes in each of 801 subjects. The population-mean CIS architecture is highly skewed (Fig. 3A): the maximum population-mean CIS is 0.00462 (node 414), and node 400 falls

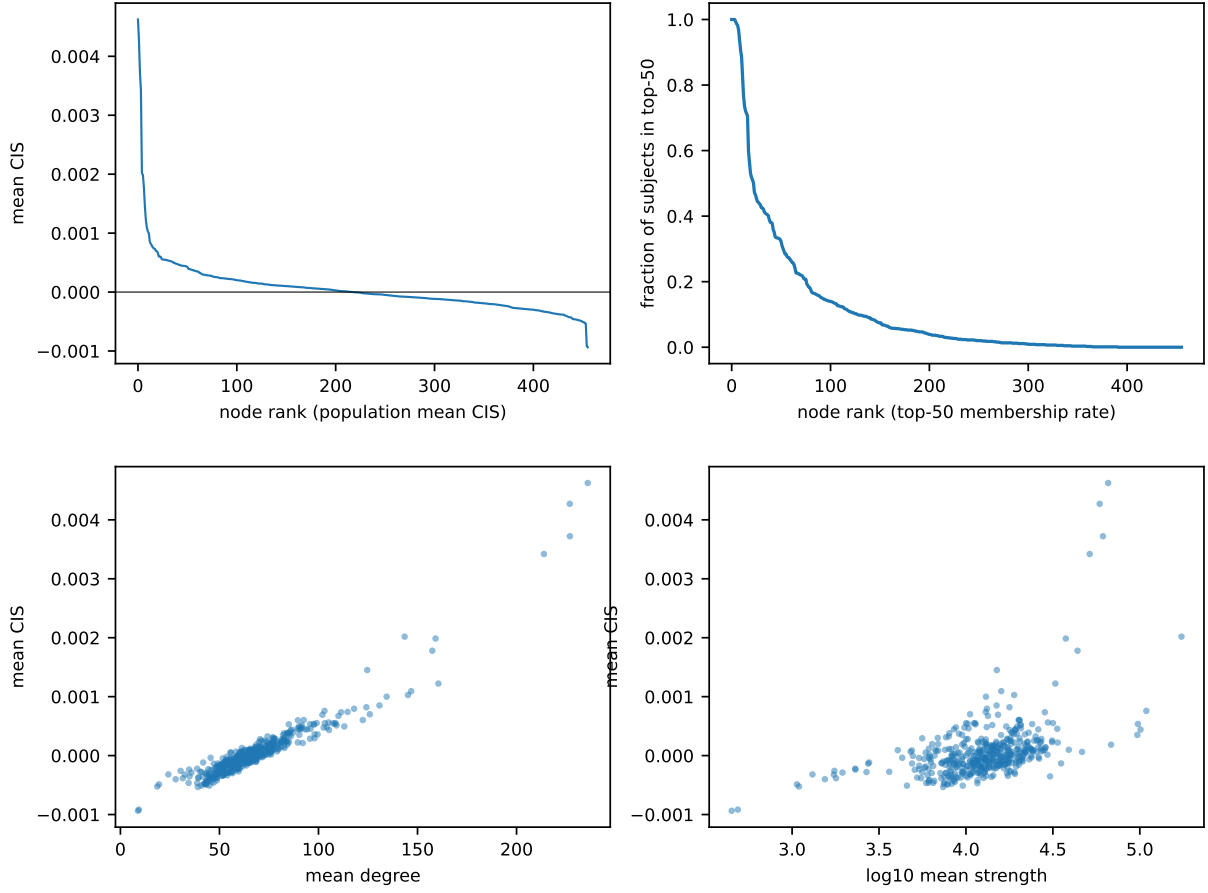


Figure 3: CIS architecture and its degree relation (801 subjects). (A) Population-mean CIS across all 456 nodes; CIS is unitless—the fraction of baseline global efficiency $E(G)$ lost on node removal—with a maximum population-mean CIS of 0.00462 (node 414). (B) CIS tracks node degree (median Spearman $\rho = 0.943$) far more closely than strength ($\rho = 0.486$), motivating the degree-controlled residual analysis of Fig. 5.

in the per-subject top 50 in 100% of subjects. Small absolute magnitudes are expected at 15% cost in a redundant graph with mean degree $\langle k \rangle = 68.2$. CIS tracks degree far more closely than strength (median Spearman $\rho = 0.943$ vs. 0.486; Fig. 3B), motivating the degree-controlled residual analysis. The highest-stability nodes are listed in supplementary Table S1.

3.2 Degree-preserving null arbitration (H1)

Top-50 CIS concentration exceeded its degree-preserving null in **200/200 subjects** (median $z_{\text{vs null}} = 16.36$; two-sided sign $p = 1.24 \times 10^{-60}$; Fig. 4B). All per-subject empirical p -values sit at the 1/101 null-resolution floor; inference therefore rests on the z statistics and the population sign test, which we state explicitly. Per-node Stouffer z against battery A is finite for 43 of 456 nodes (maximum 23.30 at node 400), with the remaining 413 nodes censored at the empirical- p floor (Fig. 4A).

3.3 Degree-independent residual (H2)

The residual (Eq. 2) is positive in **778/801 subjects** (sign $p = 2.6 \times 10^{-197}$) with median Cliff’s $\delta = 0.100$ (across-subject IQR [0.078, 0.129]; Fig. 5). Of the 36,846 individual matched comparisons, 58.8% beat the control; the median paired Wilcoxon p is 0.167 (125/801 subjects $< .05$). The residual is therefore *small in effect size but directionally near-universal across subjects*—measurable only through matched controls plus null arbitration.

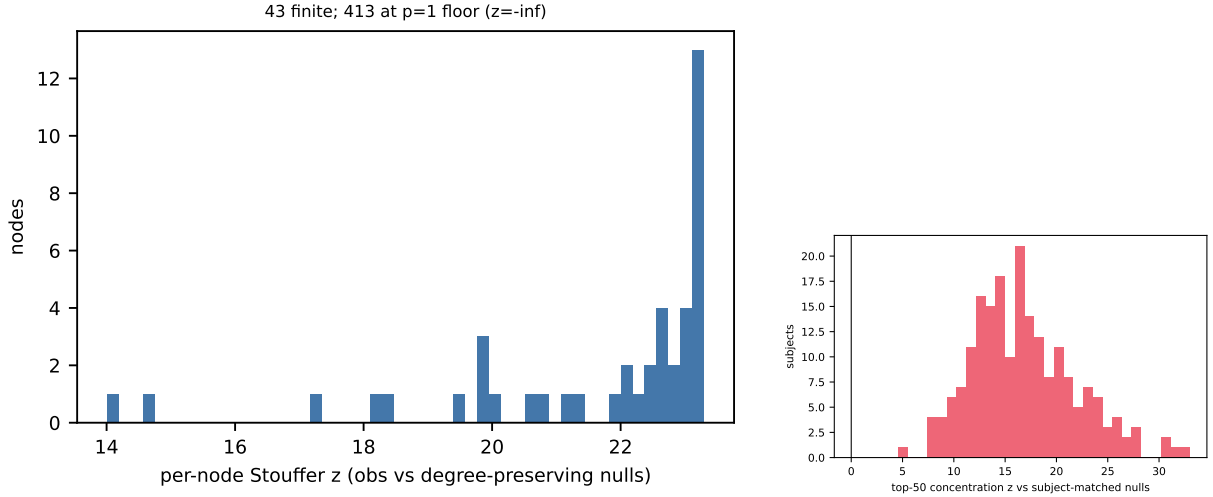


Figure 4: Degree-preserving null arbitration. (A) Per-node Stouffer z against the 100-null battery (battery A): 43 of 456 nodes carry finite z (maximum 23.30, node 400); the remaining 413 nodes sit at the add-one empirical- p floor and are explicitly counted as censored ($-\infty$), not hidden. (B) Per-subject top-50 concentration z (battery B): all 200/200 subjects lie above their null ensembles. All per-subject empirical p -values sit at the 1/101 null-resolution floor; inference therefore rests on the z statistics and the population sign test (two-sided $p = 1.24 \times 10^{-60}$).

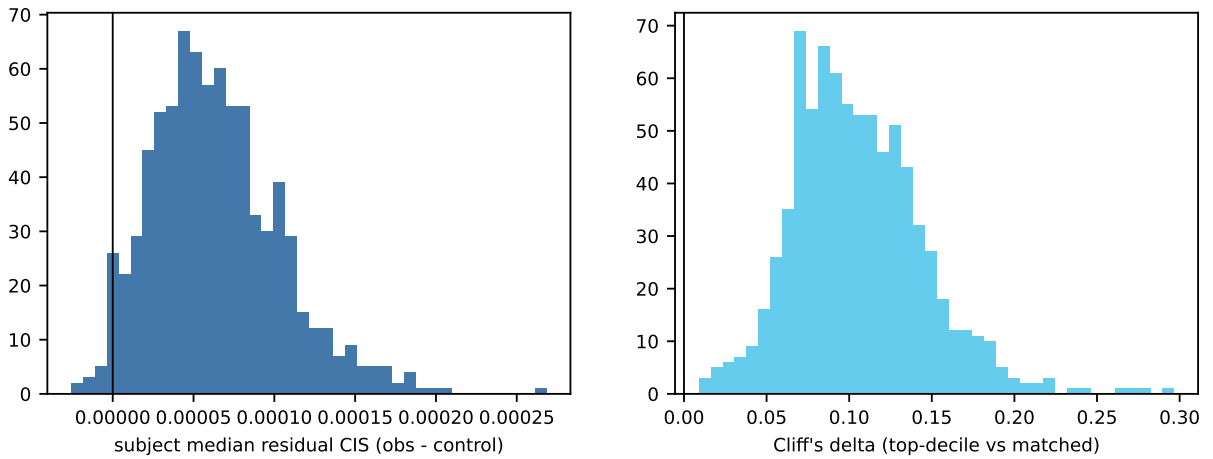


Figure 5: Degree-independent residual architecture (801 subjects). Subject-median CIS residual against degree- and strength-matched controls (36,846 pairs), and the per-subject Cliff's δ distribution: the residual is positive in 778/801 subjects (sign $p = 2.6 \times 10^{-197}$) with median $\delta = 0.100$ [IQR 0.078–0.129]; 58.8% of individual matched comparisons beat the control, and the median paired Wilcoxon p is 0.167 — small in effect size, directionally near-universal across subjects.

3.4 Anatomical distribution and the pre-registered system gate (H3/R2b)

43/456 nodes survive BH-FDR $q < .05$ (maximum Stouffer $z = 23.30$, node 400; cluster spanning nodes 400–451). Their anatomical distribution is dominated by subcortical/cerebellar tissue (39 of 43), with limbic (2) and default-mode (2) nodes (Fig. 6). The pre-registered system-enrichment gate (R2b) was, however, **negative**: visual cortex holds 1 of the top 50 (2%; $z_B = -0.38$), and the single nominal system signal (Subcortical/Cerebellar: observed 26, $z_A = 9.06$, $z_B = 2.08$, $p = .031$, 1.25 $\times$) is absent at $K = 25$ ($z_B = 0.81$) and degree-anchored. The four pre-specified sensory/attention systems show no positive degree-matched enrichment (Table 2). H3 is therefore *not supported*: no robust canonical-system-specific explanation of the residual was established.

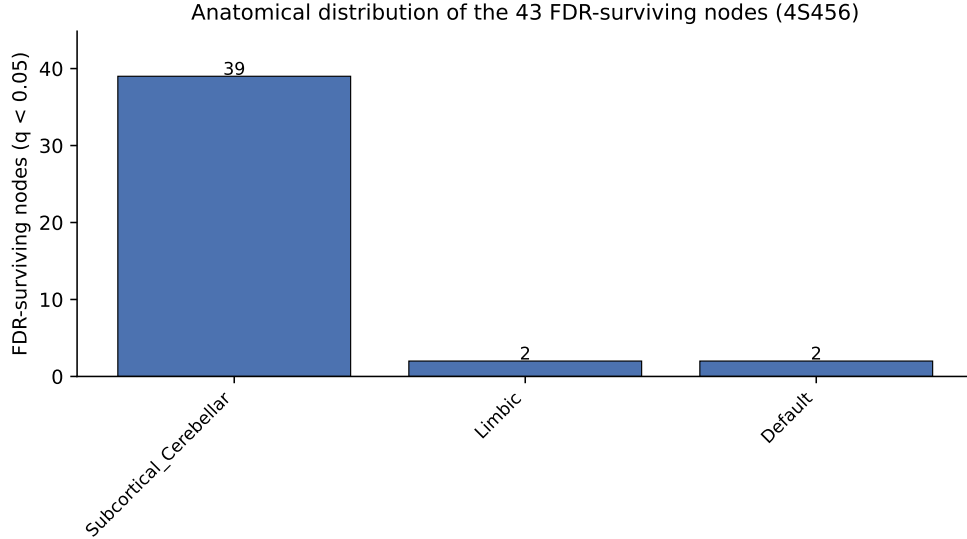


Figure 6: Anatomical distribution of the 43 FDR-surviving nodes. System counts of the 43/456 nodes surviving BH-FDR $q < .05$ (maximum Stouffer $z = 23.30$): Subcortical/Cerebellar 39, Limbic 2, Default 2. The pre-registered system-enrichment gate (R2b) itself was negative: visual cortex holds 1 of the top 50 (2%), and the single nominal system signal (Subcortical/Cerebellar, $z_B = 2.08$, $p = .031$, $1.25\times$) is degree-anchored and absent at $K = 25$. Rendered from the frozen `table_07_null_results.csv` plus the atlas label manifest; no statistics were recomputed.

Table 2: Pre-registered system-enrichment gate (R2b) at $K = 50$ for the four pre-specified sensory/attention systems: observed count in the top-50 residual nodes, node-label-shuffle control (z_A , p_A ; 10,000 permutations), and degree-matched peer-pool control (z_B , p_B ; 10,000 resamples), with control-B enrichment. **The pre-registered gate was negative:** no pre-specified sensory/visual system shows positive degree-matched enrichment (Visual: 1/50 nodes, $z_B = -0.38$). The full eight-system table, including the nominal Subcortical/Cerebellar signal ($z_B = 2.08$, $p = .031$, $1.25\times$; absent at $K = 25$ and degree-anchored), is frozen in the E05 artifacts. Source: RESOLUTION.md (verbatim).

System	Obs.	z_A	p_A	z_B	p_B	Enrich. (B)
Visual	1	-2.52	0.9995	-0.38	0.7754	0.69x
Somatomotor	2	-2.59	0.9989	-1.63	0.9813	0.38x
Dorsal attention	3	-1.03	0.9097	-0.09	0.6296	0.96x
Saliency/ventral attention	8	1.40	0.1292	-0.50	0.7524	0.86x

3.5 Robustness (E06)

All nine pre-registered alternative configurations (atlases {4S256, 4S156, Brainnetome246Ext, AAL116}; weights {sift_invnodevol, radius2_count}; costs {0.10, 0.20, 0.25}; frozen $n = 150$ cohort, nested samples) reproduce the primary architecture (Table 3; supplementary Fig. S2). The maximum like-for-like top-50-mean deviation is 0.00262 (full precision 0.0026194; atlas AAL116: 0.0038480 vs. the like-for-like primary 0.0012286 on the same cohort). *No numeric tolerance was pre-registered for E06* in PROTOCOL_FREEZE.md or STUDY_DESIGN.md; the concordance is a descriptive comparison, not a pass/fail gate, and we report the exact deviation rather than a threshold verdict. Rank stability across 100 half-splits is high (median Spearman 0.996, IQR 0.996–0.997).

3.6 Cross-scale comparison (H4)

Table 4 and Fig. 7 give the pre-registered comparators. The human residual effect size ($\delta = 0.100$) and the fly value ($\delta = 0.0979$) are concordant, and both architectures are degree-dominated. Two boundaries must be stated with the result. First, the fly δ *did not survive* its own degree-preserving null ensemble ($p = 0.109$); the cross-scale δ agreement is interesting but partly fortuitous (different units of analysis, graphs, and designs). Second, the residual’s anatomical identity diverges: visual systems hold 2% of

Table 3: E06 robustness ladder: nine pre-registered alternative configurations (frozen $n = 150$ cohort, nested samples). Each condition reproduces the primary architecture. Δ_{50} : like-for-like deviation of the top-50 mean-of-means from the like-for-like primary row (4S456, radius2_count, cost 0.15, same cohort; primary top-50 mean 0.0012286). *: maximum absolute deviation, full precision 0.0026194 (atlas AAL116: 0.0038480 vs 0.0012286). No numeric tolerance was pre-registered for E06 in PROTOCOL_FREEZE or STUDY_DESIGN; the concordance is a descriptive comparison, not a pass/fail gate. Sources: RESOLUTION.md; table_08_robustness.csv.

Condition	n	top-50 mean	Δ_{50}	Gini (med.)	Δ Gini	ρ (CIS, deg.) (med.)
cost 0.10	150	0.001564	0.000335	0.52647	0.03486	0.8978
cost 0.20	150	0.001029	-0.000200	0.47547	-0.01614	0.9791
cost 0.25	150	0.000926	-0.000302	0.45767	-0.03395	0.9950
atlas 4S256	150	0.001887	0.000658	0.50955	0.01794	0.9422
atlas 4S156	150	0.003179	0.001951	0.56178	0.07017	0.9566
atlas Brainnetome246Ext	150	0.001704	0.000476	0.45887	-0.03274	0.9463
atlas AAL116	150	0.003848	0.002619*	0.53892	0.04731	0.9447
weights sift_innodevol	150	0.001187	-0.000042	0.48316	-0.00845	0.9379
weights radius2_count	150	0.001254	0.000026	0.48526	-0.00635	0.9489

Table 4: Pre-registered cross-scale comparators (human vs. fly), normalized per the frozen discipline tags (DIRECT: same statistic and control design; NORMALIZED: scale-normalized analogue; QUALITATIVE). Absolute CIS values are never compared across species (different units, graphs, and designs). The fly δ did not survive its own degree-preserving null ensemble ($p = 0.109$). Human and fly nodes are not treated as homologous structures; the comparison operates at the level of network architecture and statistical organization. Source: table_09_cross_scale.csv (verbatim).

Comparator	Fly (FAFB v783)	Human (AOMIC-ID1000)	Tag
Effect size of the CIS shift (target vs. matched controls), Cliff's δ	Cliff's delta = 0.0979	Cliff's delta = 0.100 (median across 801 subjects; residual CIS vs degree-matched controls)	DIRECT
Null survival of the group-level CIS effect (degree-preserving ensembles)	$z = 1.21$, $p = 0.109$ (NOT survived)	battery B (E04b/E05): see NULL_MODEL_REPORT; top-50 median vs 100 degree-preserving nulls	DIRECT
Top- K enrichment vs. tested universe (control A: label shuffle)	$K=50$ visual_centrifugal: obs 21, enrich 11.7x, $z_A = 14.5$	E05 system enrichment z_A per system	NORMALIZED
Top- K enrichment vs. degree-matched expectation (control B)	$K=50$: E_B 8.0, enrich 2.63x, $z_B = 5.30$, $p = 1e-4$	E05 z_B per system (degree-matched peer pools)	DIRECT
Chokepoint catalogue depth	50 nodes catalogued; 13 A_strong_chokepoint	top-100 catalogue (human_chokepoint_catalogue.csv); A/B classing pending E05 q-values	QUALITATIVE
Visual-system share of the top 50	80% (visual_system_fraction_top50, e14_v2_summary)	2% of top-50 nodes in Vis systems	DIRECT

the human top 50 versus 80% visual-centrifugal nodes in the fly top 50, and the fly visual-centrifugal enrichment (2.63 $\times$ after degree matching, $z_B = 5.30$, $p \approx 10^{-4}$; 13/50 matched peers) has no positive human analog (R2b negative). The comparison supports an architectural statement only: *architecture replicates; anatomy does not*.

3.7 Hypothesis outcomes

H1 **supported** (200/200; $z = 16.36$; $p = 1.24 \times 10^{-60}$); H2 **supported** (778/801; $\delta = 0.100$; 43/456 FDR); H3 **not supported** (R2b negative; nominal subcortical signal degree-anchored and K-ladder-fragile); H4

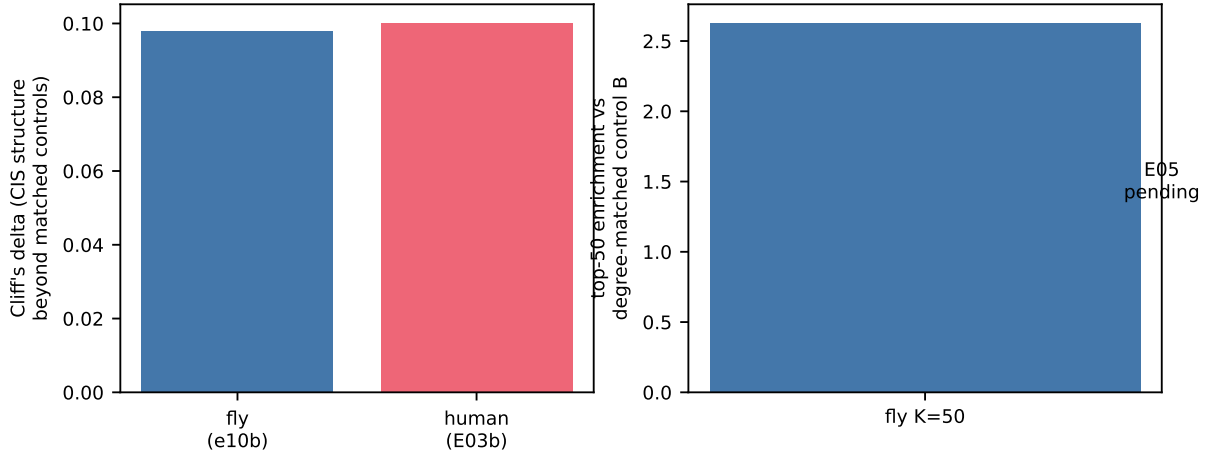


Figure 7: Cross-scale comparison (human vs. fly), normalized comparators only. Effect-size (Cliff's δ), enrichment (z_B), and concentration comparators, normalized per the cross-scale discipline tags: human $\delta = 0.100$ vs. fly $\delta = 0.0979$ (the fly value did not survive its own degree-preserving null ensemble, $p = 0.109$); human visual share of the top 50 is 2% versus 80% visual-centrifugal in the fly. Absolute CIS values are never compared across species (different units, graphs, and designs).

resolved as bounded architectural concordance with anatomical divergence (Table 4; fly δ not null-surviving). These outcomes match the frozen decision summary (table_10_decision_summary.csv).

3.8 Pre-registered design and decision framework

Every analysis in this paper was pre-registered and frozen before any outcome evaluation, in `PROTOCOL_FREEZE.md` and `STUDY_DESIGN.md`; the frozen cohort, atlas, weight variant, threshold, and the full specification of the null ensembles and the pre-specified comparison rules are recorded separately. The framework was fixed in advance and is reproduced here so that the closed nature of the inference is explicit.

The framework had three layers, each a pre-registered gate on a specific aspect of the architecture. **Layer 1 (gate on the architecture)** asks whether control impact is degree-dominated and reproducible at population scale: the top-50 CIS concentration against degree-preserving nulls, evaluated in every subject, with the two-sided population sign test as the population-level verdict. H1 is this gate. **Layer 2 (gate on the residual)** asks whether a degree-independent component of control impact exists and is reproducible: the degree/strength-matched residual, the per-subject sign of the residual, Cliff's δ with its across-subject IQR, and the count of subjects in which the paired difference is individually significant. H2 is this gate. **Layer 3 (gate on the system interpretation)** asks whether the residual is robustly system-specific and whether the architecture replicates across scales. H3 is this gate; H4 is the cross-scale comparison in the preceding subsection.

A pre-registered numeric tolerance was not set for the E06 robustness assessment. The maximum like-for-like top-50-mean deviation is 0.0026194 (full precision), with the AAL116 configuration deviating 0.0038480 versus the like-for-like primary 0.0012286 on the frozen cohort. No threshold verdict is attached to that number: it is reported as a descriptive comparison, and the concordance is a descriptive finding, not a pass/fail gate. Likewise, the H4 verdict is not a binary accept/reject. The cross-scale comparison resolves as bounded architectural concordance with divergent anatomical identity: the fly residual effect size (0.0979) did not survive its own degree-preserving null ensemble, and the human visual share of the top 50 (2%) differs from the fly visual-centrifugal share (80%). The decision summary (table_10_decision_summary.csv) records these outcomes without converting them into claims they do not support.

The design has one methodological consequence worth stating plainly. Every gate is evaluated against the frozen null ensembles that preserve the exact per-node degree sequence, so the verdicts cannot be recycled from a statistical method whose significance depends on an unverified null. This is the same

discipline that produced the preserved negative results: the fly GABA-specific hypothesis and the human R2b gate, reported here as first-class outcomes, are the load-bearing evidence that the positive architecture is read against the correct null model.

3.9 Reproducibility and audit

All reported values are frozen at the release v1.1.0 tag: no result, statistic, table, or figure was added, changed, or optimized after the release was assembled. The frozen pipeline (Python 3.13, numpy 2.4.2, scipy 1.18.0, pandas 2.3.3) and every manifest (cohort, null, figure provenance, and the decision summary) are recorded in `ANALYSIS_FREEZE.md`, `PROTOCOL_FREEZE.md`, `STUDY_DESIGN.md`, and the appendix artifacts under `10_REPORT/`. The full numeric audit is independent of any computation: `verify_final_numbers.py` is stdlib-only and recomputes every documented value from the frozen artifacts without re-running the analysis pipeline (46/46 checks). A fresh GitHub clone of the repository reproduces `main.pdf` and `supplementary.pdf` without any recompilation of the analysis, and page-render QC confirms every page is visually clean (0 blank, 0 edge-clipping, 0 density anomalies).

4 Discussion

4.1 Principal findings

Across 801 individual human structural connectomes, structural network impact decomposes into a dominant connectivity-dependent component and a smaller degree-independent residual. The degree component is not a statistical artifact. Top-50 CIS concentration exceeds degree-preserving null expectations in every tested subject (200/200), with a median $z_{vs\ null} = 16.36$ and a two-sided population sign $p = 1.24 \times 10^{-60}$. CIS tracks node degree far more closely than strength (median Spearman $\rho = 0.943$ vs. 0.486), which is exactly what a degree-dominated network should show and what null-model arbitration was designed to expose. The strong CIS–degree coupling therefore replicates, at population scale and with per-subject resolution, the well-documented degree–importance relationship reported in classical group-average studies [1, 2, 3], but now with the degree control and null arbitration that such studies lack.

The degree-independent residual is the second, subtler component. It is small in effect size but directionally near-universal: it is positive in 778/801 subjects, its median Cliff’s δ is 0.100 (across-subject IQR [0.078, 0.129]), and the median paired Wilcoxon p is 0.167. It lies near the CIS noise floor and is detectable only through the conjunction of degree-matched controls and degree-preserving null arbitration. Its per-node structure is concentrated (43 nodes; maximum Stouffer $z = 23.30$ at node 400), but its system-level interpretation is negative: the pre-registered system-specific gate (R2b) is not supported, and the scattered subcortical/cerebellar enrichment seen at nominal settings does not survive degree anchoring or the K ladder.

4.2 The degree-independent residual

The residual is small ($\delta = 0.100$; 58.8% of matched pairs; median Wilcoxon $p = 0.167$) and lies near the CIS noise floor, yet it is not noise. Its near-universality (778/801 subjects, sign $p = 2.6 \times 10^{-197}$) and its reproducibility across 9 alternative configurations (maximum like-for-like top-50-mean deviation 0.0026194) indicate a genuine, if modest, degree-independent component of control impact. It is detectable only through the conjunction of degree-matched controls and degree-preserving null arbitration; neither control layer alone would reveal it.

The per-node structure of the residual is concentrated—43 of 456 nodes carry finite Stouffer z against the battery A nulls, with the remaining 413 censored at the add-one empirical- p floor. The system-level interpretation remains negative (R2b): no pre-specified sensory or visual system is enriched, and the nominal subcortical/cerebellar signal ($z_B = 2.08$, $p = .031$) is absent at $K = 25$ and degree-anchored. This is the key negative finding of the paper. Matched-pair significance without null arbitration can mislead: the fly study’s GABA-specific hypothesis, rejected here as a preserved negative result (raw $\delta = 0.111$,

$p = 0.0011$, failing the degree-preserving null, empirical $p = 0.109$), is the canonical demonstration of exactly that failure mode.

4.3 Cross-scale architecture

A qualitatively similar two-component organization appears at both scales despite a $\sim 10^3$ -fold difference in analyzed node count and non-homologous anatomy. The relation to previous literature is bounded: classical removal analyses [1, 2, 7] are group-average without matched controls or nulls; the closest neighbour [8] uses matched-mass nulls in an ageing framing (partial overlap; full-text verified); controllability studies [5, 6] address a complementary, dynamical metric; and cross-species connectome analyses [9, 10] do not perform per-node removal rankings with degree-matched controls and degree-preserving nulls on both scales. These results motivate the hypothesis that two-component control-impact organization may recur across nervous systems of very different scale; they do not establish anatomical homology, a conserved biological mechanism, or a universal law. The cross-scale design is deliberately conservative: normalised comparators only, with explicit discipline tags (DIRECT / NORMALIZED / QUALITATIVE); absolute CIS values are never compared across species, because the graph-theoretic unit (global efficiency $E(G)$) and the parcellation are different in the two networks, and the node counts differ by roughly 10^3 -fold.

Two boundaries must be stated with the result. First, the fly δ did not survive its own degree-preserving null ensemble ($p = 0.109$), so the cross-scale concordance does not inherit fly significance; the agreement is architectural and partly fortuitous. Second, the residual’s anatomical identity diverges: visual systems hold 2% of the human top 50 versus 80% visual-centrifugal nodes in the fly top 50, and the fly visual-centrifugal enrichment ($2.63\times$ after degree matching, $z_B = 5.30$, $p \approx 10^{-4}$; 13/50 matched peers) has no positive human analog, because the human R2b gate was negative. The cross-scale comparison therefore supports an architectural statement and nothing more: architecture replicates; anatomy does not.

4.4 Negative and null findings

The following are first-class outcomes of this study, preserved as such throughout the manuscript. This is not a lapse in significance but a deliberate feature of the design: six distinct outcomes are recorded, and each is stated so that nothing that is scientifically true is concealed under a significance threshold.

(1) the pre-registered human system-specific gate (R2b) was negative; no pre-specified sensory or visual system is enriched in the residual, and the single nominal subcortical/cerebellar signal ($z_B = 2.08$, $p = .031$) is absent at $K = 25$ and degree-anchored; (2) the fly GABA-specific hypothesis was rejected—the raw matched-pair effect ($\delta = 0.111$, $p = 0.0011$) did not survive the degree-preserving null ensemble (observed $\delta = 0.098$ vs. null 0.071 ± 0.022 ; empirical $p = 0.109$; median-difference $p = 0.782$; $z = 1.21$); (3) the fly residual effect size itself is not null-surviving ($p = 0.109$); (4) battery-B per-subject empirical p -values do not resolve the tail (1/101 null-resolution floor), so inference rests on the z statistics and the population sign test, which we state explicitly; (5) the subcortical residual is not robust across the pre-registered K ladder, and its FDR cluster overlaps the same node block as the nominal enrichment, so R2a and R2b are not independent confirmations; and (6) no causal, homology, mechanism, or universal-law claim is made at any point, a discipline enforced throughout the manuscript.

We record these as negative findings rather than excise them as post hoc qualifications. The discipline is methodological as well as rhetorical: matched-pair significance without null arbitration can mislead, and the fly study’s GABA-specific hypothesis is precisely the canonical demonstration of that failure mode—a near-significant raw effect ($\delta = 0.111$, $p = 0.0011$) that dissolves entirely once the same nodes are compared under degree-preserving nulls ($z = 1.21$, empirical $p = 0.109$). The negative gate on the human side (R2b) and the rejected fly hypothesis are thus not afterthoughts; they are load-bearing parts of the architecture we are reporting. Removing them would present the positive results as uniquely compelling, when in fact the positive results rest on the very null arbitration that the negatives demonstrate is necessary.

4.5 Limitations

This is an observational, secondary analysis of a single acquisition and pipeline, and a series of explicit limitations bound the claims made in what follows.

(i) The data are undirected connectomes derived from a single MRI acquisition and pipeline; the AOMIC-ID1000 derivative used here does not provide directed variants, so the account is inherently undirected, and a directed analysis would be a different study.

(ii) The results are atlas- and threshold-dependent in detail: the 4S456 parcellation, the sift radius2 count weight, and the 15% proportional threshold were chosen a priori in the frozen protocol, and each is a decision that shapes the network that is then analysed. The architecture, however, is robust across the 9-configuration ladder, which is itself a finding about the stability of the degree-dependent decomposition rather than a weakness of the specific configuration.

(iii) The residual is near the CIS noise floor, which is precisely why the two-layer control design is required: either degree- or strength-matched controls alone would conflate the degree-independent component with degree-dominated artifacts.

(iv) Per-subject empirical p -values are limited by the 100-null resolution, which is why inference for the per-subject battery rests on the z statistics and the population sign test, with the null-resolution floor stated explicitly.

(v) The subcortical residual is not robust across the pre-registered K ladder, and its FDR cluster overlaps the same node block as the nominal enrichment, so R2a and R2b are not independent confirmations; the two gates do not multiply-support each other.

(vi) The literature audit cannot exclude occupation of the analysed combination in sources not surfaced by the queries used, and the environment manifest for the frozen runs has a disclosed provenance gap; both are disclosed and left open.

(vii) No anatomical homology mapping, functional equivalence, causal inference, or universal-law claim is made or supported, and the disciplined restraint in the language is itself a feature of the reported design.

4.6 Conclusion

Top-50 control-impact concentration exceeds degree-preserving null expectations in every tested human subject; a small degree-independent residual is present in 778/801 subjects and is not robustly system-specific; and the cross-scale comparison with the fly connectome shows concordant effect-size architecture with divergent anatomical identity. Architecture replicates; anatomy does not. The finding is hypothesis-generating, and the two-layer arbitration design (degree-matched controls plus degree-preserving nulls) is the methodological prerequisite for testing it on future connectomes. This design is not an unnecessary complication; it is the mechanism by which a degree-independent residual can be reported without being mistaken for degree-driven artifacts. On future connectomes of different scale, the same two-layer architecture, the same degree-matched and null-arbitrated quantification, and the same pre-registered gate structure should be applied before any system-specific claim is entertained.

The negative outcomes reported here are not residuals to be cleaned up. They are part of the finding. The pre-registered system-specific gate was negative, the fly GABA-specific hypothesis was rejected, and the fly residual effect size did not survive its own null ensemble. A cross-scale comparison that reports its negative gates with the same candor as its positive architecture is a comparison with nothing to hide. Architecture replicates; anatomy does not. The replication is architectural: a dominant degree component plus a small, near-universal, degree-independent residual. The divergence of anatomy is the limit of that replication, and it is where the next generation of comparative connectome work should begin.

5 Supplementary description

All supplementary material is provided as typeset PDF and source alongside this manuscript and is referenced from the main text by its figure or table label. The supplementary package is self-contained and is

reproduced faithfully from the frozen artifacts: every supplementary figure and table is a rendering of, or a table exported from, the frozen data and manifests. The following items are included.

Supplementary Figure S1 (`fig02_population_qc.pdf`): the QC flag distributions for the 900-subject acquisition (99 subjects carry flags: 94 isolated-node, 5 strength-IQR; retained, never deleted) with the 801-subject QC-pass cohort; cohort selection was frozen before any CIS computation.

Supplementary Figure S2 (`fig07_robustness.pdf`): the 9-configuration robustness ladder (atlases {4S256, 4S156, Brainnetome246Ext, AAL116}, weights {`sift_invnodevol`, `radius2_count`}, costs {0.10, 0.20, 0.25}), with the maximum like-for-like top-50-mean deviation 0.00262 (atlas AAL116; no numeric tolerance was pre-registered for E06 — descriptive comparison).

Supplementary Figure S3 (`fig09_conceptual.pdf`): the conceptual schematic of the two-layer architecture (dominant connectivity-dependent component plus smaller degree-independent residual, arbitrated by matched controls and degree-preserving nulls); schematic only, no data plotted.

Supplementary Tables S1–S2: the top-50 stability node list (Table S1) and the QC summary table (Table S2), both exported from the frozen manifests.

Supplementary tables are generated by `TABLE_SOURCES/ generate_tables.py` (fail-loud; all frozen-value assertions pass). The supplementary source, tables, and figures recompile with `tectonic supplementary.tex` from the frozen artifacts.

6 Data and code availability

Data: AOMIC-ID1000 (Zenodo 19796783; CC-BY-4.0) [11]. Fly reference: frozen release v1.0.0 artifacts (read-only; FAFB v783, CC BY-NC 4.0). Raw data are not redistributed by this project; third-party datasets retain their original licensing and usage terms. All human-side code, manifests, validation reports, and append-only logs are available at https://github.com/harsha-varadhan-2006/humanbrain_cross_species_cis; every number traces to an artifact (`FINAL_REPRODUCIBILITY_AUDIT.md`), and the full numeric audit runs without any recomputation via `verify_final_numbers.py` (stdlib-only; 46/46 checks).

Author contributions (CRediT)

H.V.M.: Conceptualization; Methodology; Software; Validation; Formal analysis; Investigation; Data curation; Writing — original draft; Writing — review & editing; Visualization; Supervision; Project administration. Funding acquisition: not applicable — no external funding was received or sought.

Conflict of interest

The author declares no competing interests. The project used an AI coding agent (Codebuff) for computational assistance under author supervision—a methodological disclosure, not a competing interest; all scientific judgment and responsibility remain with the author (repository research logs).

Ethics statement

This study is a secondary analysis of publicly distributed, already-published data (AOMIC-ID1000, Zenodo 19796783, CC-BY-4.0; FAFB v783, CC BY-NC 4.0). No new human data were collected; per the author’s determination of 2026-09-26, no IRB approval or exemption is required for this secondary use.

Acknowledgment of limitations of reproduction

The frozen analysis environment (Python 3.13, numpy 2.4.2, scipy 1.18.0, pandas 2.3.3) is recorded in `ANALYSIS_FREEZE.md`; the environment manifest for the original runs has a disclosed provenance gap.

All reported values are independently recomputable from frozen artifacts via the repository audit script.

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